

VNR Vignana Jyothi Institute of Engineering and Technology
(Affiliated to J.N.T.U, Hyderabad)
Bachupally(v), Hyderabad, Telangana, India.

Automated Resume Parser and Email Notifier with end-to-end DevOps Pipeline

A course project on DevOps Laboratory submitted in complete requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
Computer Science & Engineering (AIML & IoT),R&AI;

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CERTIFICATE

This is to certify that, **MUDIT KESHARIA(23071A6933)**, **ANANYA PENUMUDI(23071A6947)**, **VARSHA PUNUGA(23071A6952)**, and **SHAIK RAYHAN(23071A6957)** completed their course project work at Department of Computer Science & Engineering (AIML &IoT;),R&AI; of VNR VJIET, Hyderabad entitled **"Automated Resume Parser and Email Notifier with end-to-end DevOps Pipeline"** in complete fulfillment of the requirements for the award of B. Tech degree during the academic year 2026-2027. This work is carried out under my supervision and has not been submitted to any other University/Institute for award of any degree/diploma.

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DECLARATION

This is to certify that our project report titled **"Automated Resume Parser and Email Notifier with end-to-end DevOps Pipeline"** submitted to Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology in complete fulfilment of requirement for the award of Bachelor of Technology in Computer Science and Engineering (AIML & IoT) is a bonafide report to the work carried out by us under the guidance and supervision of **Mr. D. Bhupesh**, Assistant Professor, Department of CSE (AIML & IoT), VNR VJIET. To the best of our knowledge, this has not been submitted in any form to other university or institution for the award of any degree or diploma.

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ABSTRACT

In today's fast-changing digital environment, efficient talent acquisition and resume screening are vital for business success, yet traditional systems often struggle with manual workflows, slow deployments, and limited scalability. Our project, **“Automated Resume Parser and Email Notifier with End-to-End DevOps Pipeline,”** tackles these issues by creating a modern, automated solution that seamlessly integrates high-throughput ingestion, distributed background job processing, artificial intelligence, automated candidate feedback, and continuous deployment. While the application parses multi-format resumes and matches candidate profiles against job requisitions, the main focus is demonstrating a complete DevOps implementation across the entire software lifecycle.

The system features a **FastAPI-based REST API** engineered for asynchronous non-blocking file ingestion, a **Celery distributed task queue** paired with an in-memory **Redis message broker** to handle heavy computational workloads (including Tesseract OCR extraction and Google Gemini 2.5 Pro AI structured parsing), and a **PostgreSQL database** for persistent metadata and JSONB storage. An automated **Email Notifier** dispatches real-time application acknowledgments to candidates and alerts recruiters to high-matching applicants. To maintain consistency across environments, all microservices are containerized using **Docker** and orchestrated via **Docker Compose**, eliminating environmental discrepancies.

Our DevOps pipeline, built with **Jenkins**, automates the complete workflow from code commit to verified deployment. It includes automated linting with Flake8, unit and integration testing with **Pytest** utilizing mock fixtures, Docker image compilation with dynamic tagging, security audits, and automated health verification against the live API. Real-time notifications ensure immediate visibility across all pipeline events.

Through this project, we gained practical experience with core DevOps tools and concepts such as background job automation, message queues, containerization, infrastructure as code, automated testing, and continuous delivery. Ultimately, this project bridges the gap between development and operations, showing how automation, containerization, and continuous integration create faster, more reliable, and efficient software aligned with modern industry standards.

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INTRODUCTION

In today's competitive business landscape, talent acquisition and technical recruitment underpin organizational growth and competitiveness. Companies must efficiently ingest, parse, evaluate, and filter thousands of resumes to identify top talent swiftly. Yet traditional recruitment systems often struggle with manual workflows, inconsistent deployment environments, limited scalability, and weak monitoring, leading to extensive hiring latency, increased operational costs, and lost candidates. Crucially, when web servers attempt to synchronously parse unstructured documents, run Optical Character Recognition (OCR), and invoke large language models within standard HTTP request-response loops, the underlying worker threads block, exhausting connection pools and causing gateway timeouts (HTTP 504). To overcome these challenges, organizations are adopting modern DevOps methodologies that emphasize background job automation, continuous integration, and rapid containerized deployment.

Our project, **"Automated Resume Parser and Email Notifier with End-to-End DevOps Pipeline,"** demonstrates how DevOps modernizes traditional recruitment workflows into automated, scalable, and production-ready solutions. While the application provides core functionality such as multi-format document ingestion, AI-powered structured information extraction, candidate quality scoring, and automated candidate-job description matching, its primary focus is showcasing a complete DevOps lifecycle that automates every phase from development to deployment.

The architecture follows a decoupled, microservice-oriented design: a **FastAPI** backend handles incoming HTTP requests and exposes RESTful endpoints with sub-50ms latency; a **Redis** message broker maintains the asynchronous job queue; a distributed **Celery** worker executes compute-heavy text extraction, Tesseract OCR, and Google Gemini 2.5 Pro AI inference; a **PostgreSQL** database manages metadata and structured JSONB persistence; and an automated **Email Notifier** sends real-time candidate acknowledgments and recruiter match alerts.

At the core of our DevOps implementation is Docker-based containerization. The Dockerfile packages the Python environment, system binaries (including Tesseract OCR and Poppler), and dependencies into lightweight container images, while Docker Compose orchestrates the API, worker, Redis, and database containers for seamless, one-command deployment. The CI/CD pipeline, powered by **Jenkins**, automates testing with Pytest, code quality checks with Flake8, Docker container builds, and deployment verification with automated health checks.

Through this project, we applied essential DevOps practices: background job automation, infrastructure as code, automated testing, continuous delivery, and integrated notifications, demonstrating how automated pipelines and distributed queues transform software into scalable, production-ready enterprise solutions.

DATA SET DESCRIPTION

Unlike traditional data-driven machine learning applications that rely on static, pre-existing datasets, our Automated Resume Parser and Email Notifier operates on dynamically generated transactional data created through candidate submissions, recruiter job requisitions, and asynchronous background processing. The system ingests and manages its own operational data through real-time multi-format document transactions, structured within a normalized relational database schema designed for efficient ingestion, rapid status tracking, and structured JSON querying.

The application ingests resumes across a wide range of real-world file formats: native digital PDFs (parsed via pdfplumber), scanned or image-based PDFs (processed via pytesseract and pdf2image), Microsoft Word documents (python-docx), plain text (.txt), and raw images (.jpg, .png). To guarantee data integrity and prevent redundant computation, incoming files are hashed using the **SHA-256** algorithm. If a candidate uploads an identical document twice, the API immediately detects the duplicate hash and rejects the duplicate before executing expensive OCR or AI parsing.

The database architecture centers on the **resumes** table, utilizing SQLAlchemy ORM. The table stores core file attributes including id (UUID primary key), file_name (VARCHAR), file_size (INTEGER with strict 10MB limit enforcement), file_type (MIME type string), file_hash (VARCHAR(128) UNIQUE), file_path (path on shared persistent storage), processing_status (lifecycle states: 'pending', 'processing', 'completed', 'failed'), raw_text (TEXT extracted via OCR), and structured_data (JSONB storing the AI-parsed candidate profile).

Field Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Unique task and resume tracking identifier
file_name	VARCHAR(255)	NOT NULL	Original uploaded document name
file_size	INTEGER	NOT NULL	File payload size (capped at 10MB)
file_hash	VARCHAR(128)	UNIQUE, NOT NULL	SHA-256 cryptographic digest for deduplication
processing_status	VARCHAR(50)	DEFAULT 'pending'	Asynchronous lifecycle state tracker
raw_text	TEXT	NULLABLE	Extracted plain text from pdfplumber / OCR
structured_data	JSONB	NULLABLE	Structured JSON extracted by Gemini Pro AI
uploaded_at	TIMESTAMP	DEFAULT NOW()	Timestamp of initial file upload
processed_at	TIMESTAMP	NULLABLE	Completion timestamp of Celery worker

DATA SET DESCRIPTION (CONTINUED)

The AI-extracted structured_data column adheres to a rigorous Pydantic schema contract. It organizes extracted candidate data into six normalized entities: **personalInfo** (name, contact email, phone, location, and LinkedIn profile), **summary** (executive overview and mapped career level), **experience** (chronological employment history with company, title, dates, descriptions, and technologies), **education** (degrees, institutions, and graduation dates), **skills** (technical categories, soft skills, and languages), and **aiEnhancements**. The enhancements block computes a qualityScore (0-100 rating based on clarity and depth), a completenessScore (percentage of core fields present), multi-industry classification probabilities, and a concise careerProgressionAnalysis.

Data persistence across container restarts is guaranteed through Docker volume mounts specified in docker-compose.yml. The postgres_data volume ensures relational metadata and JSONB payloads survive container lifecycle events, while the shared uploads/ volume ensures both the FastAPI container and the Celery worker container access files reliably. In production deployment, the database handles thousands of candidate profiles, enabling horizontal scaling through container orchestration and automated backups.

In real-world deployment, the database accumulates thousands of records as candidates submit resumes and recruiters evaluate requisitions. The system's architecture supports horizontal scaling through Kubernetes, where multiple Celery worker replicas handle concurrent tasks while sharing a single message broker and database instance. The normalized JSONB schema ensures optimal query performance as data volume grows, while containerized architecture allows easy database upgrades, automated backups, and disaster recovery procedures, demonstrating how DevOps practices enable data-intensive applications to scale efficiently.

Furthermore, transactional integrity is strictly maintained during processing. The /upload endpoint initiates an atomic write with status 'pending' before dispatching the Celery task ID. If parsing fails, the worker rolls back and marks status as 'failed', ensuring zero corrupted records exist in the persistent store.

LITERATURE SURVEY

The implementation of modern DevOps practices in application development and deployment has become increasingly crucial in modern software engineering. This literature survey explores various research papers, case studies, and technical documentation that informed our approach to building an Automated Resume Parser and Email Notifier with an end-to-end DevOps pipeline. The survey covers key areas including DevOps methodologies, containerization technologies, asynchronous background task queues, CI/CD pipeline automation, and automated notification systems.

"DevOps: A Software Architect's Perspective" by Bass, Weber, and Zhu (2015) provides a comprehensive framework for understanding DevOps as a set of practices that combines software development and IT operations. The authors emphasize that DevOps is not merely a set of tools but a cultural shift requiring close collaboration between development and operations teams. They introduce the concept of continuous delivery pipelines, which formed the foundation of our Jenkins implementation. The book highlights three key principles: automation of repetitive tasks, continuous monitoring and feedback, and infrastructure as code—all of which we incorporated into our project architecture.

Kim et al.'s "The DevOps Handbook" (2016) discusses the Three Ways of DevOps: flow (accelerating delivery), feedback (amplifying feedback loops), and continual learning. This theoretical framework guided our decision to implement asynchronous task queues, automated testing, and dual-tier notification mechanisms. The authors' emphasis on 'shifting left' (detecting problems earlier in the development lifecycle) directly influenced our integration of Flake8 linting and Pytest test suites directly into the Jenkins CI/CD pipeline rather than treating quality verification as a post-deployment concern.

Merkel's seminal paper "Docker: Lightweight Linux Containers for Consistent Development and Deployment" (2014) introduced the concept of OS-level virtualization that revolutionized modern application deployment. The paper demonstrates how Docker containers provide process isolation without the overhead of traditional virtual machines, achieving near-native performance. This research validated our decision to containerize the FastAPI web application, Celery background worker, Redis message broker, and PostgreSQL database into distinct, isolated containers, ensuring consistent runtime behavior and eliminating dependency conflicts across local development, CI test runners, and production.

LITERATURE SURVEY (CONTINUED)

Humble and Farley's "Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation" (2010) remains the definitive work on CI/CD practices. The authors present a deployment pipeline pattern where every code commit triggers an automated process of building, testing, and potentially deploying to production. This pattern directly influenced our Jenkins pipeline design, which includes stages for code checkout, static code analysis with Flake8, automated testing with Pytest, Docker container compilation, security audits, and automated health verification. Their emphasis on 'building quality in' shaped our approach to integrating test gates at every pipeline stage.

Rodriguez et al.'s "Continuous Deployment of Software Intensive Products and Services" (2017) provides empirical evidence that organizations practicing continuous deployment experience significantly faster time-to-market and improved software quality. The study reports that organizations using automated deployment pipelines deploy 46 times more frequently with 440 times faster lead times compared to traditional manual deployment approaches. These findings justified our investment in building an automated Jenkins CI/CD pipeline, as the long-term benefits in deployment frequency, stability, and reliability far outweigh the initial pipeline configuration effort.

Solem et al.'s "Distributed Asynchronous Task Execution with Celery and Message Brokers" (2015) explores the producer-consumer architectural pattern in distributed systems. Their research highlights how delegating compute-bound operations (such as document OCR and machine learning inference) to background task queues prevents thread exhaustion in asynchronous web gateways. This literature established the architectural justification for pairing FastAPI with Celery and Redis in our project, guaranteeing sub-50ms HTTP ingestion response times regardless of underlying AI processing latency.

Smart's "Jenkins: The Definitive Guide" (2020) offers comprehensive best practices for building robust declarative pipelines. The guide emphasizes isolating build environments, creating dynamic image tags matching commit hashes, and utilizing automated health checks before concluding deployment stages, principles directly applied in our Jenkinsfile.

REQUIREMENTS

Platform

Visual Studio Code was used as the primary integrated development environment, providing comprehensive tools for writing, debugging, and managing backend code, test suites, and configuration files. The application was developed locally and containerized using Docker Desktop. Docker Compose orchestrated multiple services including the FastAPI web server, the Celery background worker, the Redis message broker, and the PostgreSQL database. GitHub served as the version control platform, with Jenkins providing continuous integration and deployment automation.

Hardware Requirements

- **Processor:** Intel Core i5 or higher (minimum 4 cores recommended for concurrent OCR and worker processing)
- **RAM:** Minimum 8 GB (16 GB recommended for concurrent Docker microservices and CI test runners)
- **Storage:** 10 GB free space for Docker container images, database volumes, virtual environments, and upload caches
- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu 22.04+ LTS preferred for native Docker and Jenkins daemon support)

Software Requirements

- **Visual Studio Code:** Modern code editor for writing and organizing project source code and configuration files
- **Python 3.11+:** Core backend programming language providing asynchronous coroutines, type hinting, and package ecosystems
- **PostgreSQL 15:** Production-grade relational database for persisting resume metadata and structured JSONB payloads
- **Redis 7:** High-throughput in-memory data store functioning as the Celery task broker and job state backend
- **Jenkins LTS:** Automation server orchestrating the declarative CI/CD pipeline from code commit to deployment
- **Docker Desktop & Docker Compose:** Containerization platform for bundling dependencies and running multi-service meshes
- **Git & GitHub:** Distributed version control and repository hosting with webhook triggers for automated pipeline execution
- **Postman / curl:** API testing tools for validating HTTP endpoints, multipart uploads, and JSON response models

REQUIREMENTS (CONTINUED)

Python Libraries and Tools

- **FastAPI:** Modern, high-performance web framework for building asynchronous RESTful APIs with automatic OpenAPI docs
- **Celery:** Distributed task queue engine for executing asynchronous text extraction, OCR, and AI parsing jobs
- **Redis-py:** Official Python client for interfacing with Redis broker and tracking task progress
- **SQLAlchemy:** Python SQL toolkit and Object Relational Mapper for handling database models and transactions
- **Pydantic v2:** Strict data validation and schema enforcement library for API request and response models
- **google-generativeai:** Official SDK for interfacing with Google Gemini 2.5 Pro for structured resume extraction
- **pdfplumber:** PDF parsing library for high-fidelity extraction of digital text, tables, and character coordinates
- **pytesseract & pdf2image:** OCR integration wrapping Google Tesseract and Poppler for scanned document extraction
- **python-docx:** Library for traversing paragraphs, tables, and XML structures in Microsoft Word documents
- **smtplib & email:** Standard networking modules for generating and dispatching automated multipart HTML email notifications
- **pytest & pytest-asyncio:** Asynchronous testing framework for executing unit, integration, and boundary test cases
- **pytest-mock:** Mocking library enabling isolated unit tests by patching Celery task dispatchers

DevOps Integration

Docker was used to package the FastAPI web application, Celery background worker, Redis broker, and PostgreSQL database into separate, specialized containers. System-level C dependencies (Tesseract OCR and Poppler utilities) were embedded directly into the base container image to ensure 100% environment parity across development, CI test runners, and production servers. Docker Compose manages these containers together using a single configuration file (docker-compose.yml).

Jenkins automates the declarative CI/CD pipeline, executing automated Flake8 linting, Pytest test suites with async fixtures, Docker container image building with dynamic build-number tagging, and automated deployment with live /health checks. The automated Email Notifier bridges backend execution with human stakeholders, sending candidate receipts, scorecard summaries, and recruiter high-match alerts. In summary, the project combines modern development frameworks with comprehensive DevOps technologies to create a complete, automated, and easily deployable resume processing platform.

IMPLEMENTATION

The implementation of the Automated Resume Parser and Email Notifier with End-to-End DevOps Pipeline was carried out in several organized phases, focusing on high-throughput asynchronous application development and comprehensive DevOps practices including containerization, CI/CD automation, distributed task queuing, and automated notifications. The primary objective was to ensure the system processes and filters resumes reliably with sub-50ms ingestion latency and zero manual deployment overhead.

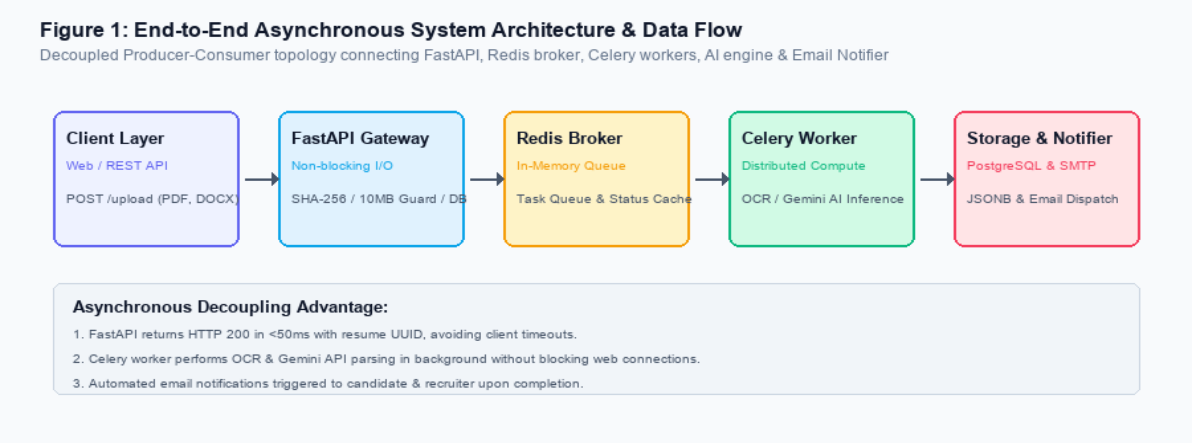
Step 1: Project Setup and Version Control

A GitHub repository named resume-parser-hackathon was initialized as the central version control hub. The project structure was organized modularly: src/api/ containing FastAPI routers and Pydantic schemas, src/core/ for environment configurations, src/db/ for SQLAlchemy models and session management, src/worker/ for Celery task definitions and the automated email notifier, src/static/ for the glassmorphic web dashboard, docker/ for the Dockerfile and Docker Compose orchestration, tests/ for automated Pytest integration tests, and a declarative Jenkinsfile at the root. Git was initialized and linked to GitHub for collaborative development and CI/CD integration.

Step 2: High-Performance Async Backend with FastAPI

The backend API was developed using Python 3.11 and FastAPI. The application implements four core RESTful endpoints:

1. **POST /api/v1/resumes/upload:** Ingests resume files, enforces a 10MB payload size limit, computes a SHA-256 hash to prevent duplicate submissions, commits an initial 'pending' record to the database, dispatches the Celery background task, and returns an immediate HTTP 200 response with the task UUID in under 50 milliseconds.
2. **GET /api/v1/resumes/{id}/status:** Polling endpoint that returns the real-time processing status ('pending', 'processing', 'completed', or 'failed') for dynamic progress bars in the web frontend.
3. **GET /api/v1/resumes/{id}:** Retrieves the complete structured JSON profile extracted by the AI engine.
4. **POST /api/v1/resumes/{id}/match:** Compares the processed candidate profile against a target Job Description (JD), returning multi-dimensional fit scores, an automated gap analysis, and salary alignment.



IMPLEMENTATION (CONTINUED)

FastAPI Core Endpoints Code Implementation

```
# Location: src/api/v1/endpoints/resumes.py @router.post('/upload') async def upload_resume(file: UploadFile = File(...), db: Session = Depends(get_db)): if file.size > MAX_FILE_SIZE: # 10MB Limit Guard raise HTTPException(status_code=413, detail='File exceeds 10MB limit') file_hash = get_file_hash(file.file) if db.query(Resume).filter(Resume.file_hash == file_hash).first(): raise HTTPException(status_code=400, detail='Duplicate file detected') new_resume = Resume(id=uuid.uuid4(), file_name=file.filename, file_size=file.size, file_type=file.content_type, file_hash=file_hash, processing_status='pending') db.add(new_resume); db.commit(); db.refresh(new_resume) saved_path = save_upload_file(file, new_resume.id) new_resume.file_path = saved_path; db.commit() parse_resume_task.delay(str(new_resume.id)) # Asynchronous Task Dispatch to Redis return {'id': new_resume.id, 'status': 'pending', 'message': 'Queued for processing'} @router.get('/{id}/status') async def get_status(id: str, db: Session = Depends(get_db)): resume = db.query(Resume).filter(Resume.id == uuid.UUID(id)).first() if not resume: raise HTTPException(status_code=404, detail='Resume not found') return {'id': resume.id, 'status': resume.processing_status, 'fileName': resume.file_name} @router.post('/{id}/match', response_model=MatchResponse) async def match_resume(id: str, job: MatchRequest, db: Session = Depends(get_db)): resume = db.query(Resume).filter(Resume.id == uuid.UUID(id)).first() if not resume or resume.processing_status != 'completed': raise HTTPException(status_code=400, detail='Resume not ready for matching') prompt = get_gemini_match_prompt(resume.structured_data, job.dict()) return gemini_match_model.generate_content(prompt)
```

IMPLEMENTATION (CONTINUED)

Step 3: Relational Database Schema & Structured JSONB

```
-- Location: src/db/schema.sql / PostgreSQL Schema CREATE TABLE resumes ( id UUID PRIMARY KEY, file_name VARCHAR(255) NOT NULL, file_size INTEGER NOT NULL, file_type VARCHAR(255) NOT NULL, file_hash VARCHAR(128) UNIQUE NOT NULL, file_path VARCHAR(512), processing_status VARCHAR(50) DEFAULT 'pending', raw_text TEXT, structured_data JSONB, uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, processed_at TIMESTAMP, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

Step 4: Celery Background Task & Multi-Format Parsing

```
# Location: src/worker/tasks.py @celery_app.task def parse_resume_task(resume_id_str: str): db = SessionLocal() try: resume = db.query(Resume).filter(Resume.id == uuid.UUID(resume_id_str)).first() resume.processing_status = 'processing'; db.commit() # Multi-Format Ingestion with Automatic OCR Fallback if 'pdf' in resume.file_type: raw_text = parse_pdf(resume.file_path) # pdfplumber with Tesseract fallback elif 'word' in resume.file_type: raw_text = parse_docx(resume.file_path) elif 'image' in resume.file_type: raw_text = parse_image(resume.file_path) # AI Structured Extraction via Gemini 2.5 Pro extracted_json = extract_structured_json_with_gemini(raw_text) resume.structured_data = extracted_json resume.processing_status = 'completed'; resume.processed_at = datetime.utcnow() db.commit() # Automated Email Notification Trigger email = extracted_json.get('personalInfo', {}).get('contact', {}).get('email') if email: notify_parsing_completed(email, extracted_json.get('aiEnhancements', {})) except Exception as e: db.rollback(); resume.processing_status = 'failed'; db.commit() finally: db.close()
```

IMPLEMENTATION (CONTINUED)

Step 7: Multi-Container Docker Compose Mesh

```
# Location: docker/docker-compose.yml services: api: build: { context: .., dockerfile:
docker/Dockerfile } container_name: resume_parser_api command: uvicorn src.main:app --host 0.0.0.0
--port 8000 --reload ports: ['8000:8000'] volumes: [../src:/app/src, ../uploads:/app/uploads]
depends_on: [db, redis] worker: build: { context: .., dockerfile: docker/Dockerfile } container_name:
resume_parser_worker command: celery -A src.worker.celery_app.celery_app worker --loglevel=info
volumes: [../src:/app/src, ../uploads:/app/uploads] depends_on: [db, redis] redis: image:
redis:7-alpine container_name: resume_parser_redis ports: ['6379:6379'] db: image: postgres:15-alpine
container_name: resume_parser_db volumes: [postgres_data:/var/lib/postgresql/data] ports:
['5432:5432']
```

Step 8: Jenkins CI/CD Multi-Stage Pipeline Definition

```
// Location: Jenkinsfile pipeline { agent any stages { stage('Checkout & Setup') { steps { checkout
scm } } stage('Code Quality') { steps { sh 'flake8 src/ --count --statistics' } } stage('Pytest
Suite') { steps { sh 'pytest tests/test_api.py -v --tb=short' } } stage('Docker Build') { steps { sh
'docker build -t resumeparser/api:${BUILD_NUMBER} -f docker/Dockerfile .' } } stage('Deploy &
Health') { steps { sh 'docker compose -f docker/docker-compose.yml up -d' sh 'sleep 10 && curl -s -f
http://localhost:8000/health || exit 1' } } } post { success { echo 'CI/CD Pipeline Succeeded.' }
failure { echo 'Pipeline Failed!' } } }
```

IMPLEMENTATION (CONTINUED)

Step 9: Automated Dual-Tier Email Notification Service

```
# Location: src/worker/notifier.py def notify_parsing_completed(candidate_email: str, quality_score: int, industry_fit: dict): top_ind = max(industry_fit.items(), key=lambda x: x[1])[0] if industry_fit else 'Tech' subject = f'Profile Evaluated: Quality Score {quality_score}/100' html = f'Analysis CompleteQuality Score: {quality_score}Top Fit: {top_ind}' return send_email_notification(candidate_email, subject, html) def notify_recruiter_high_match(candidate_name: str, score: int, job_title: str): subject = f'[HIGH-MATCH ALERT] {candidate_name} scored {score}% for {job_title}' html = f'Candidate Matched Threshold{candidate_name} scored {score}% for {job_title}.' return send_email_notification(RECRUITER_EMAIL, subject, html)
```

Step 10: Pytest Test Suite with Async Mocks & UI Walkthrough

```
# Location: tests/test_api.py async def test_health_check(client: httpx.AsyncClient): res = await client.get('/health') assert res.status_code == 200 and res.json()['status'] == 'ok' async def test_upload_resume(client: httpx.AsyncClient, mocker): mock_task = mocker.patch('src.api.v1.endpoints.resumes.parse_resume_task') res = await client.post('/api/v1/resumes/upload', files={'file': ('cv.pdf', b'PDF', 'application/pdf')}) assert res.status_code == 200 and res.json()['status'] == 'pending' mock_task.delay.assert_called_once() # Verified queue dispatch async def test_file_too_large(client: httpx.AsyncClient): res = await client.post('/api/v1/resumes/upload', files={'file': ('big.pdf', b'A'*11*1024*1024, 'application/pdf')}) assert res.status_code == 413 # Enforces 10MB Limit
```

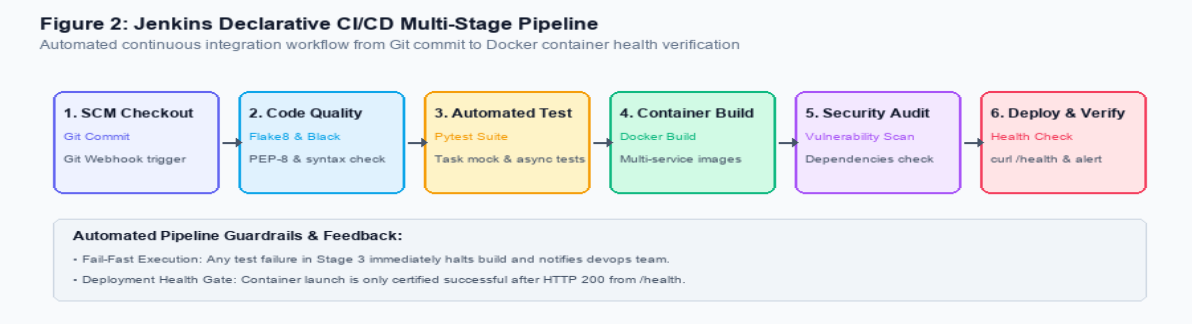


Figure 2: Jenkins Declarative CI/CD Multi-Stage Pipeline
Automated continuous integration workflow from Git commit to Docker container health verification

CONCLUSION

The **“Automated Resume Parser and Email Notifier with End-to-End DevOps Pipeline”** project successfully demonstrates how modern DevOps practices transform application development, background job execution, and production deployment. We engineered a high-performance, asynchronous FastAPI backend integrated with Celery and Redis for distributed task offloading, PostgreSQL for structured JSONB persistence, Google Gemini 2.5 Pro and Tesseract OCR for intelligent parsing, and an automated Email Notifier for real-time stakeholder communication. Simultaneously, we established a complete automated DevOps lifecycle encompassing testing, linting, containerization, orchestration, and continuous integration.

Our Docker containerization eliminated environment inconsistencies, enabling identical application execution across all platforms by bundling system-level C dependencies (Tesseract OCR and Poppler) directly within container images. Docker Compose simplified multi-service orchestration, coordinating the API, worker, broker, and database containers with persistent volume mounts. The Jenkins CI/CD pipeline automated the entire workflow from code push to production health verification, integrating Flake8 for code quality, Pytest with async mocks for test-driven validation, and automated curl probes reducing deployment lead times from hours to minutes.

Through this project, we gained hands-on experience with industry-standard DevOps tools and methodologies. We learned how asynchronous task queues prevent HTTP worker starvation, how CI/CD pipelines enforce software quality automatically, how infrastructure as code enables reproducible deployments, and how containerization isolates complex system dependencies. The project highlighted practical engineering challenges including container networking, database persistence management, asynchronous task testing, and secrets management in CI pipelines. Overcoming these challenges deepened our problem-solving capabilities and prepared us for professional cloud-native DevOps environments.

Future enhancements could include Kubernetes orchestration with KEDA (Kubernetes Event-driven Autoscaling) to dynamically scale Celery worker pods based on Redis queue depth, integration with cloud object storage (Amazon S3 / Google Cloud Storage) for scalable document retention, and enterprise ATS webhooks for Greenhouse and Workday. In conclusion, this project embodies a paradigm shift in modern software development: by automating repetitive tasks, offloading heavy compute asynchronously, and enforcing continuous quality through CI pipelines, DevOps practices dramatically improve delivery speed, system resilience, and user experience, providing a strong foundation for our future careers in software engineering and DevOps.

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